Rebika Parajuli

Pokhara, Nepal | rebikaparajuli@gmail.com | +977 9767655682
rebikaparajuli.com.np | GitHub | LinkedIn | Hashnode

SUMMARY

Computer Engineering student at IOE Paschimanchal Campus with a strong interest in AI and cybersecurity. Built a phishing-detection browser extension, a RAG-based legal assistant prototype and an automated AI video pipeline, and practise security skills through picoCTF and TryHackMe challenges with published write-ups.

EDUCATION

Bachelor of Engineering in Computer Engineering, IOE Paschimanchal Campus, Pokhara, Nepal | Expected graduation: Nov 2028
High School, Gandaki Boarding School, Pokhara, Nepal | 2022–2024

TECHNICAL SKILLS

Cybersecurity: Linux command line, networking fundamentals, Netcat, OSINT, CTF challenges (picoCTF, TryHackMe), VirusTotal API
AI & machine learning: PyTorch, NumPy, Pandas, RAG, Tesseract OCR, local LLMs with Ollama
Programming: Python, C++, C, JavaScript, Bash
Web & backend: FastAPI, Flask, React, Streamlit, HTML, CSS, SQLite
Other tools: Faster-Whisper, Kokoro TTS, Manim, FFmpeg, Git, GitHub

PROJECTS

Phishing Finder: Firefox Security Extension | JavaScript, Python (FastAPI), VirusTotal API
Individual project, built with ChatGPT's help | Code: GitHub

- Checks each visited URL against VirusTotal through a local FastAPI backend and warns the user only when a URL looks suspicious or high-risk. Install from GitHub (not on the Firefox Add-ons store).

CyberChat: Automatic Cybersecurity Videos | Python, Ollama (Qwen3 8B), Kokoro TTS, Manim, Faster-Whisper, FFmpeg
Individual project; code written with AI help (ChatGPT early stages, Gemini later stages) | YouTube: CyberChat channel | Article: Medium write-up

- Pipeline for educational cybersecurity videos (tool explainers, incident case studies, concept deep dives): a local AI model writes the script, Kokoro narrates, Manim animates, Faster-Whisper adds captions, FFmpeg joins everything, and a YouTube upload step publishes it.
- I designed the pipeline, verified each stage and revised the design as problems came up: moved script writing from Mistral to Qwen3 8B, added safeguards to limit uploads to once per day and prevent concurrent jobs (plus a failure log), and fixed a render failure caused by a GPU library (cuBLAS) that could not be loaded.
- Status: tested stage by stage; a full scheduled run is not yet confirmed end to end.

MIRA: Civic Legal Assistant (Nepal Land Law) | Python, Tesseract OCR, RAG
Team project, CodeFest 2026 (provincial). My part: backend and AI (OCR and RAG) | Code: GitHub

- Reads uploaded documents such as municipal notices with Tesseract OCR and summarises them in simpler language. A chatbot answers questions by searching an index of Nepal's Lands Act first (RAG); the prototype covers this one law only. Status: working prototype, not deployed yet.

Flower Recognition Using Deep Learning | PyTorch, ResNet9, Streamlit
Learning project, built by following a YouTube tutorial | Code: GitHub | Live demo: Streamlit app

- Trained a ResNet9 model on a Kaggle dataset to tell 5 flower types apart; my run reached 78% accuracy after 20 epochs. Deployed as a Streamlit web app.

Other projects: BAT (Python, UPower; built with ChatGPT's help), a Linux battery health monitor showing charge, cycle count and power data (GitHub); **FollowPulse** (React, Flask, Selenium), a dashboard that shows mutual followers from an Instagram export (GitHub, demo).

SECURITY PRACTICE & WRITING

- **5 CTF write-ups** on Hashnode: picoCTF keygenme-py (reverse engineering), First Grep, Mod 26 (ROT13), What's a net cat?; TryHackMe Missing Person (OSINT)
- **Medium article:** CyberChat: Building an Automated Cybersecurity Video Production Pipeline with Local AI

EXPERIENCE, HACKATHONS & CERTIFICATES

AWS Cloud Club, IOE Paschimanchal Campus | *Marketing & Social Media Head / Outreach & Branding Member* | May 2025 – May 2026: led social media, branding and outreach for technical events.
US–Nepal Hackathon 2026 (36-hour online), team member on Within, an AI-based student support project (certificate); **CodeFest 2026** (provincial), team member on MIRA.
6-month AI Bootcamp (2025–2026), i-CES, IOE Paschimanchal Campus (certificate); **Silver Medalist**, GBS Maths League 2023 (certificate); **Bronze Medalist**, BNKS Maths League 2024 (certificate).

Languages: English, Nepali, Hindi